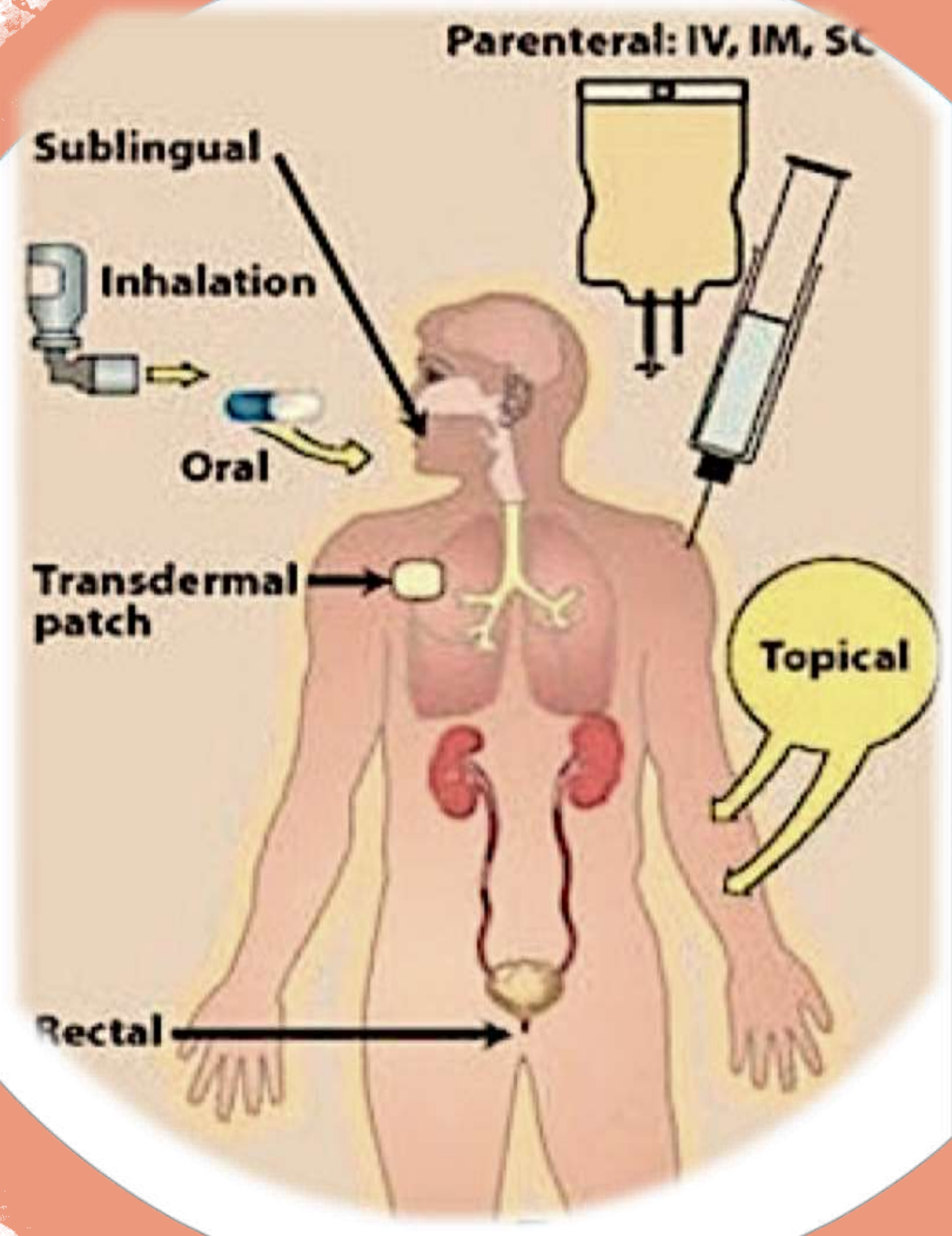
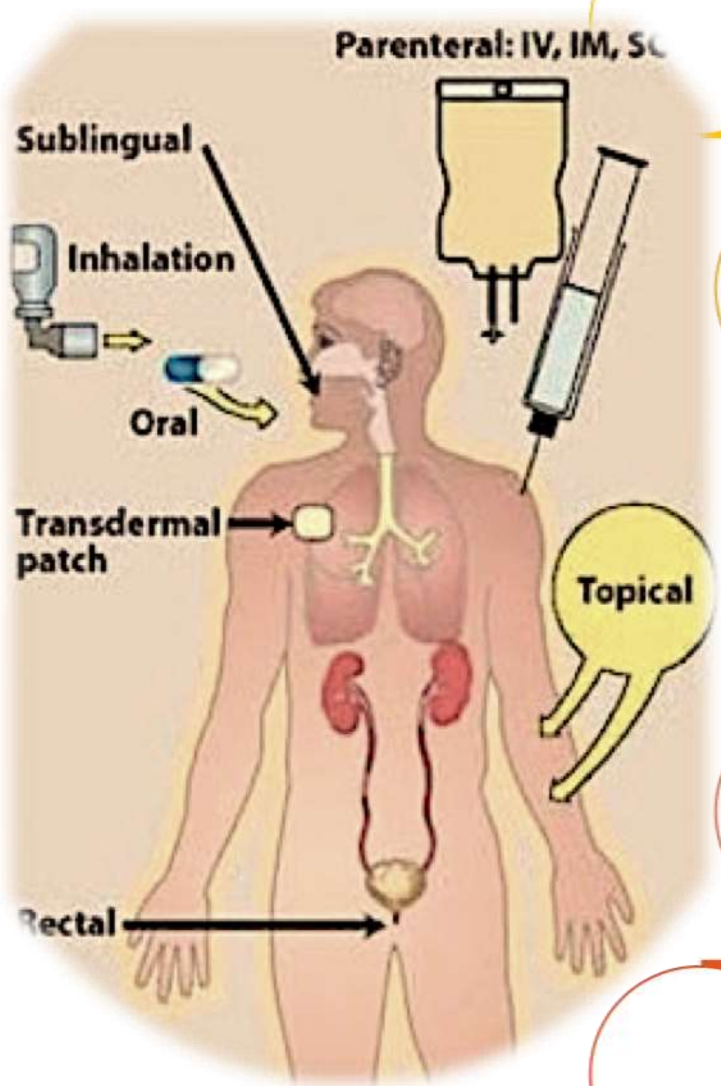
Routes of Drug Administration



Clinical Pharmacology Department,
Faculty of Medicine,
KAU

Objectives

Upon completion of this lesson, the student will be able to:



- Describe various routes of drug administration.
- Discuss pharmacokinetic implications of various routes of administration.
- Plan the route of the drug depending on clinical condition of the patient.
- Describe suitable sites for every type of injection.
- Discuss the relevant advantages and disadvantages of various routes of drug administration.

Routes of drug administration

Routes of drug administration

- **Enteral**
  - Oral
  - Sublingual
  - Rectal
- **Parenteral**
  - Intravenous
  - Intramuscular
  - Subcutaneous
  - Intradermal
  - Transdermal
- **Topical**
  - Skin topical
  - Intranasal
  - Ophthalmic
  - Otic drops
  - Inhalation
  - Vaginal
  - Other

Entero = GI track (intestine)
Para = beside

Routes of drug administration

- Parenteral
  - Intravenous
  - Intramuscular
  - Subcutaneous
  - Intradermal
  - Transdermal

Intravenous administration.

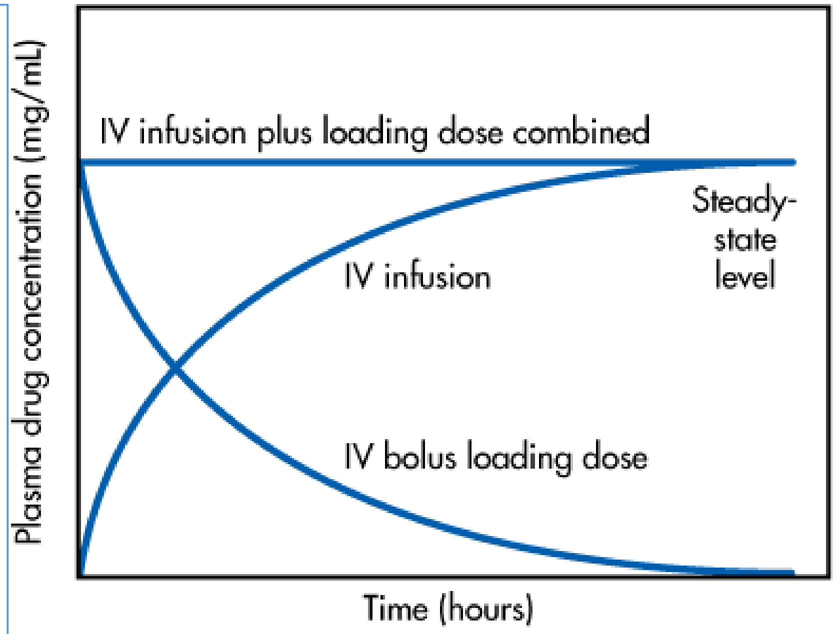
This route delivers 100% of the drug directly into the circulation (**bioavailability** = 100%).

1-Bolus.

- The entire dose is administered rapidly, producing a high plasma concentration and, usually, the most rapid effect.
- Injection volume is limited only by the size of the patient (i.e., plasma volume) and the properties (i.e., toxicities) of the drug or fluid administered.

2- Infusion.

- The drug is administered at a constant rate.
- Injection volume is unlimited.
- I.V route of administration carries the highest risk of adversely affecting vital functions.



Source: Shargel L, Wu-Pong S, Yu ABC: *Applied Biopharmaceutics & Pharmacokinetics, 6th Edition*: www.accesspharmacy.com

Routes of drug administration

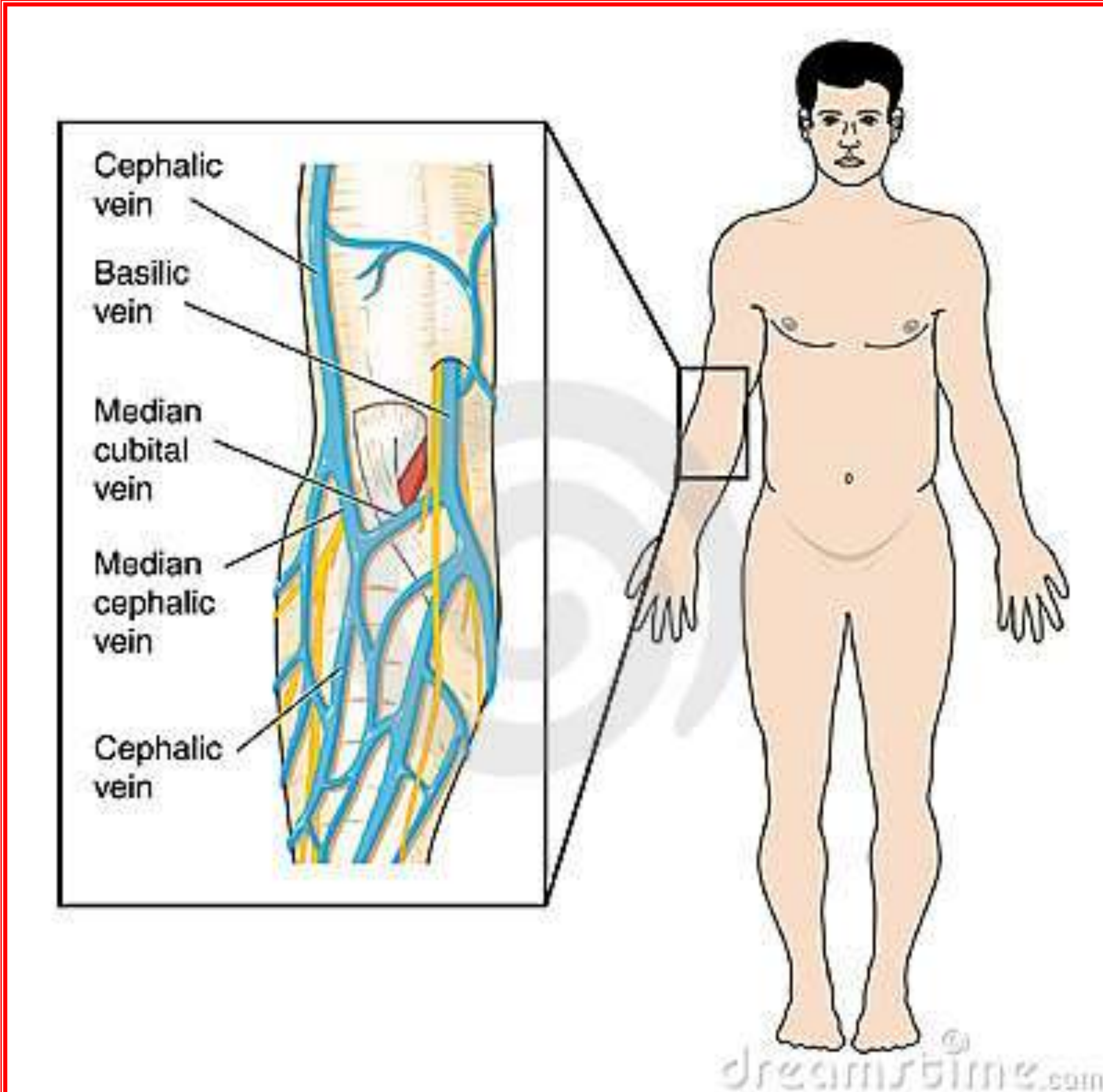
- Parenteral
  - Intravenous
  - Intramuscular
  - Subcutaneous
  - Intradermal
  - Transdermal

Intravenous administration.

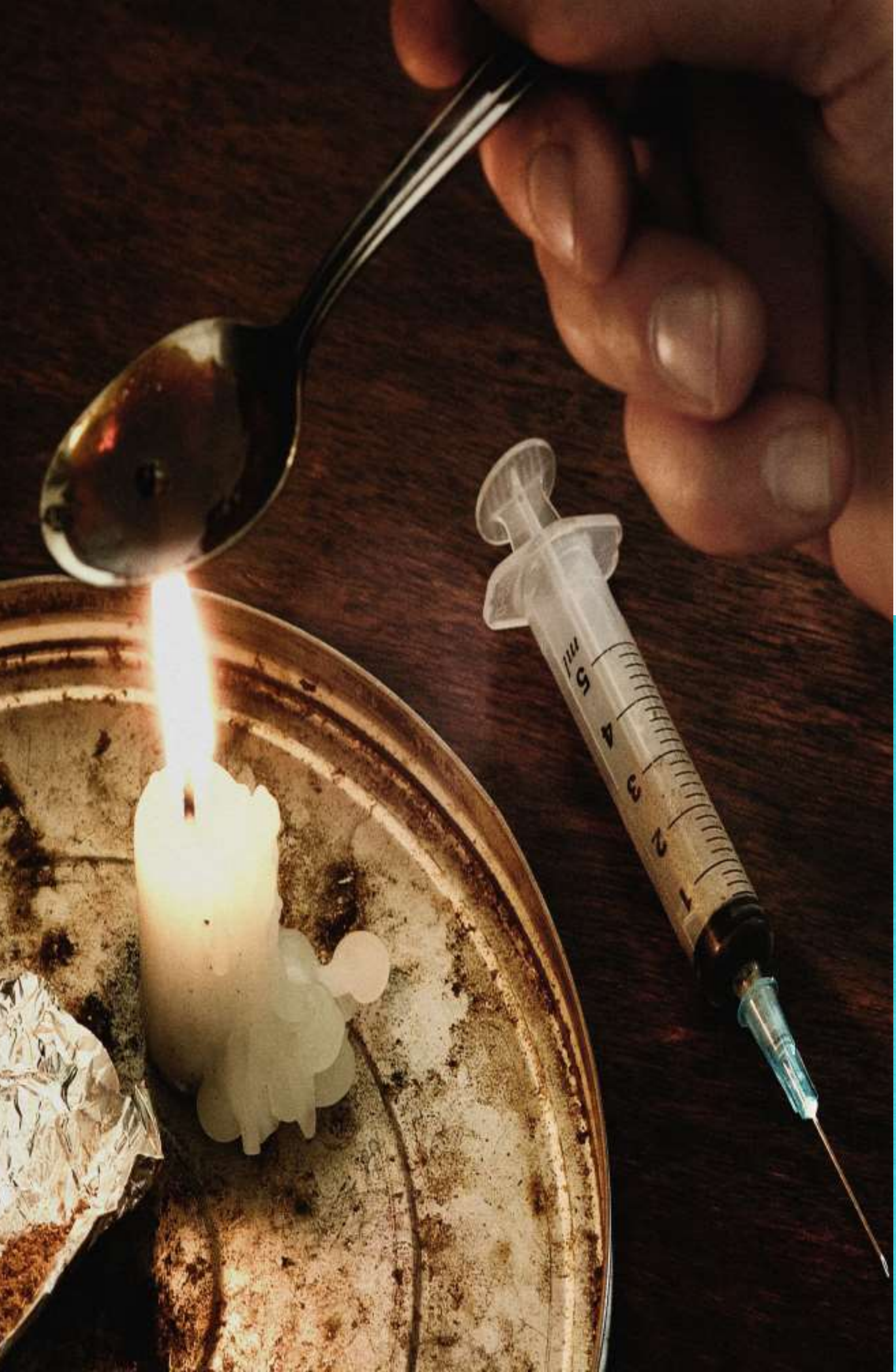
Advantages

Disadvantages

Advantages	Disadvantages
• Rapid action. • Can be employed in unconscious/ uncooperative patients. • Suitable for emergencies. • Proper dose adjustment. • Suitable for large volumes. • By-pass liver metabolism. • Suitable for irritant drugs.	• Not suitable for suspensions or oily solutions. • Once administered difficult to eliminate from the body. • Needs strict sterility. • Expensive. • Inconvenient (painful) for the patient.


Cephalic vein
Basilic vein
Median cubital vein
Median cephalic vein
Cephalic vein
dreamstime.com

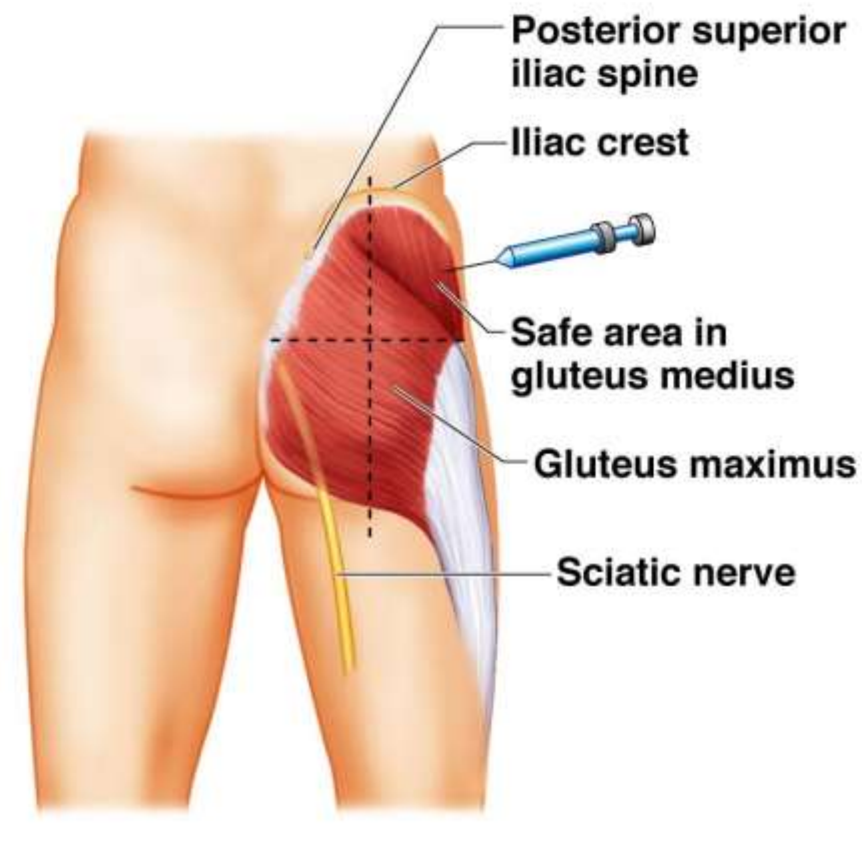
HEALING VEIN DAMAGE FROM IV DRUG USE



- Parenteral
  - Intravenous
  - Intramuscular
  - Subcutaneous
  - Intradermal
  - Transdermal

Intramuscular administration.

The drug is administered into muscle tissue, usually deep muscle, such as the gluteus or deltoid (bioavailability < 100%).



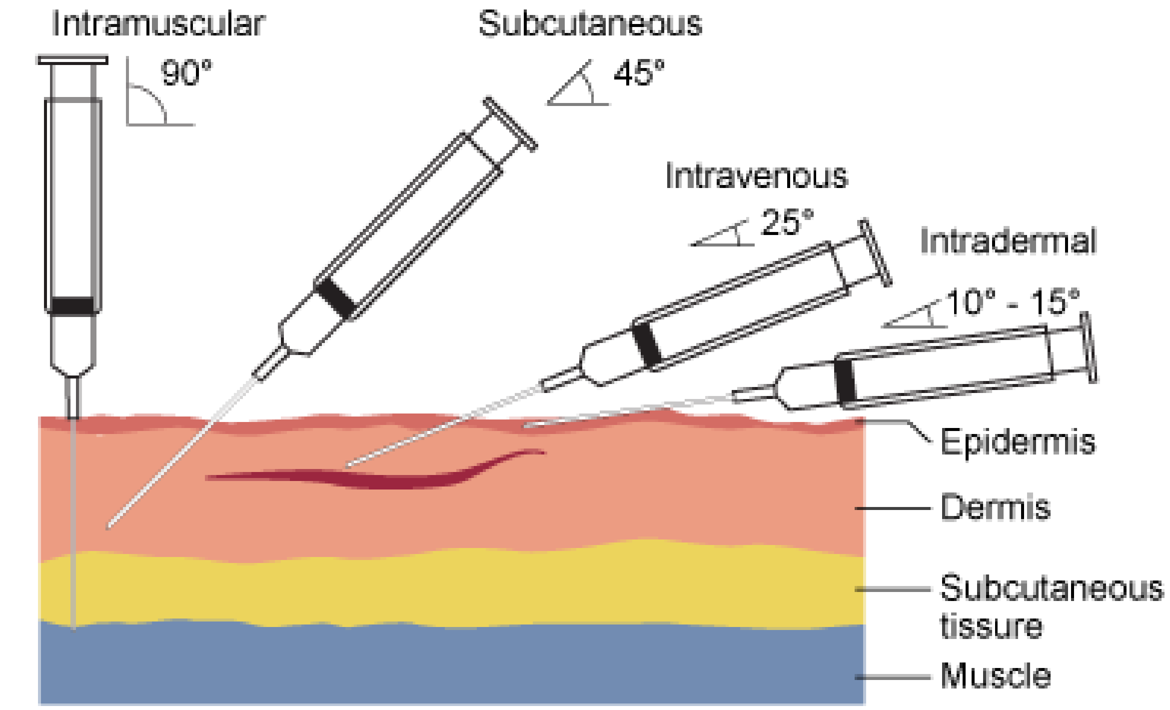
(b)

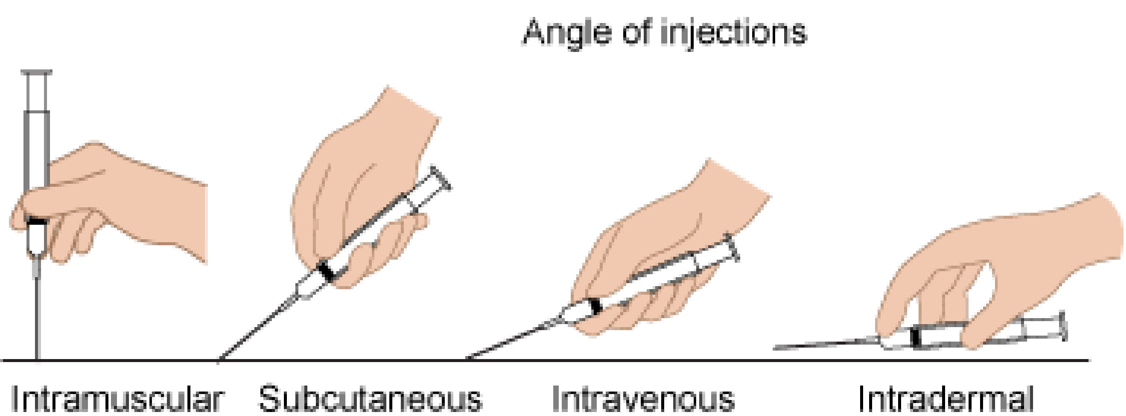
Advantages

Disadvantages

Advantages	Disadvantages
• - Suitable for suspensions or oily solutions. • - Suitable for irritant drugs. • - Suitable for moderate volumes.	- Not suitable for patient on anticoagulant therapy.(risk of hematoma) -Expensive.

The maximum injection volume is approximately 2—5 ml.


Intramuscular
90°
Subcutaneous
45°
Intravenous
25°
Intradermal
10° - 15°
Epidermis
Dermis
Subcutaneous tissue
Muscle


Angle of injections
Intramuscular
Subcutaneous
Intravenous
Intradermal


90°
Subcutaneous layer
Muscle layer


SAN TAN Allergy & Asthma

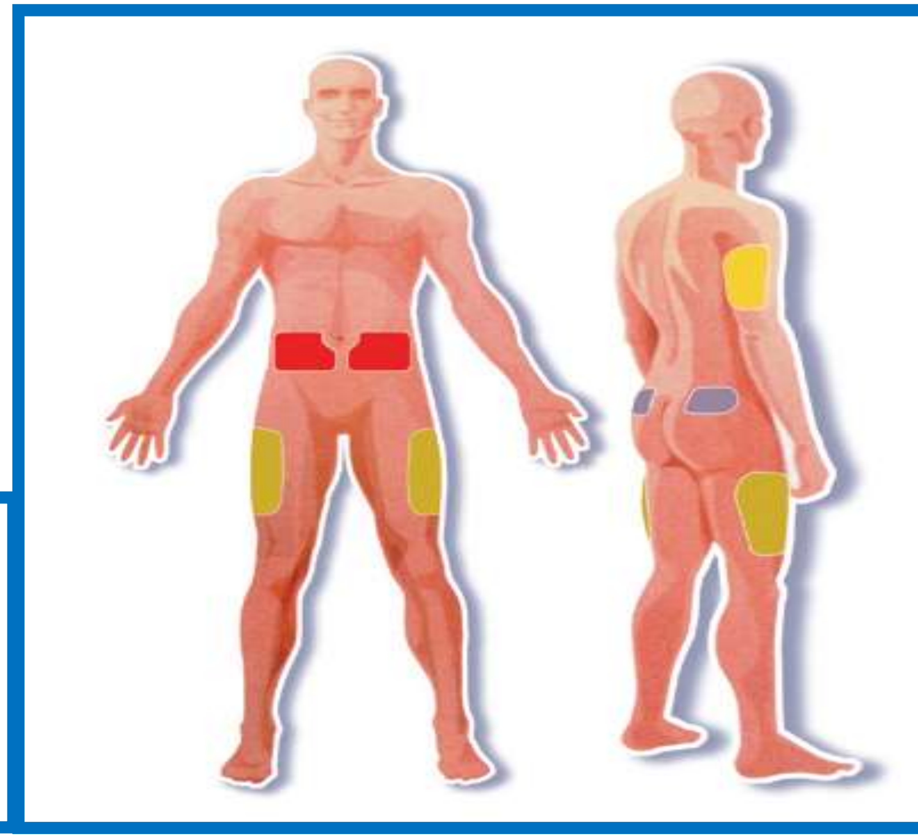
- Parenteral
  - Intravenous
  - Intramuscular
  - Subcutaneous
  - Intradermal
  - Transdermal

Subcutaneous administration.

1- Subcutaneous injection.

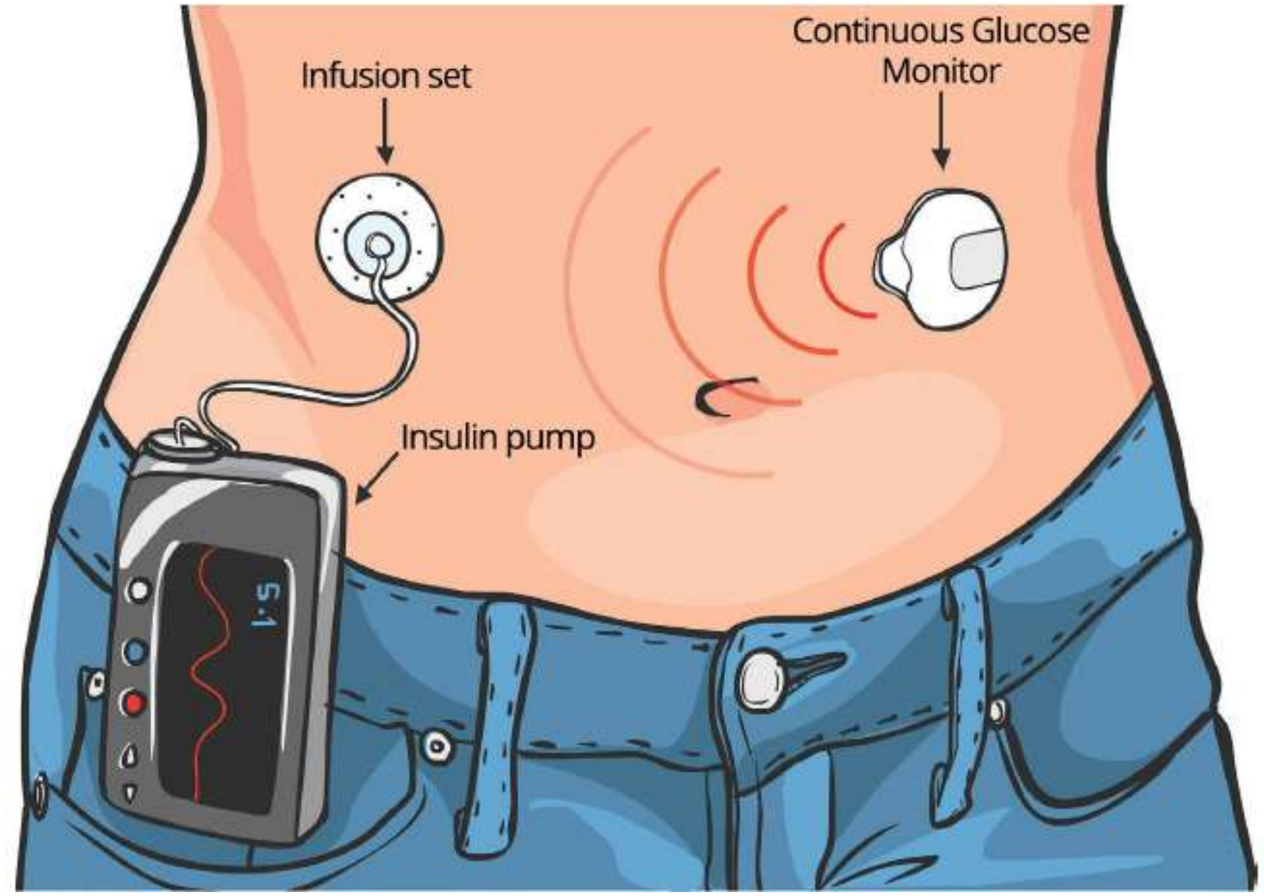
The drug is injected into subcutaneous tissue (bioavailability <100%).

Sites of Subcutaneous Injection



Advantages	Disadvantages
• Prolonged duration of action (longer than with IV), owing to slower release into the circulation. • Suitable for suspensions or oily solutions. • S.C pellets can be used to give an action for 3-6 months.	• Slow onset of action. • Not suitable for irritant drugs. • Not suitable for large volumes (The maximum injection volume is approximately **1 ml**.)


300 250 200 150 100 50

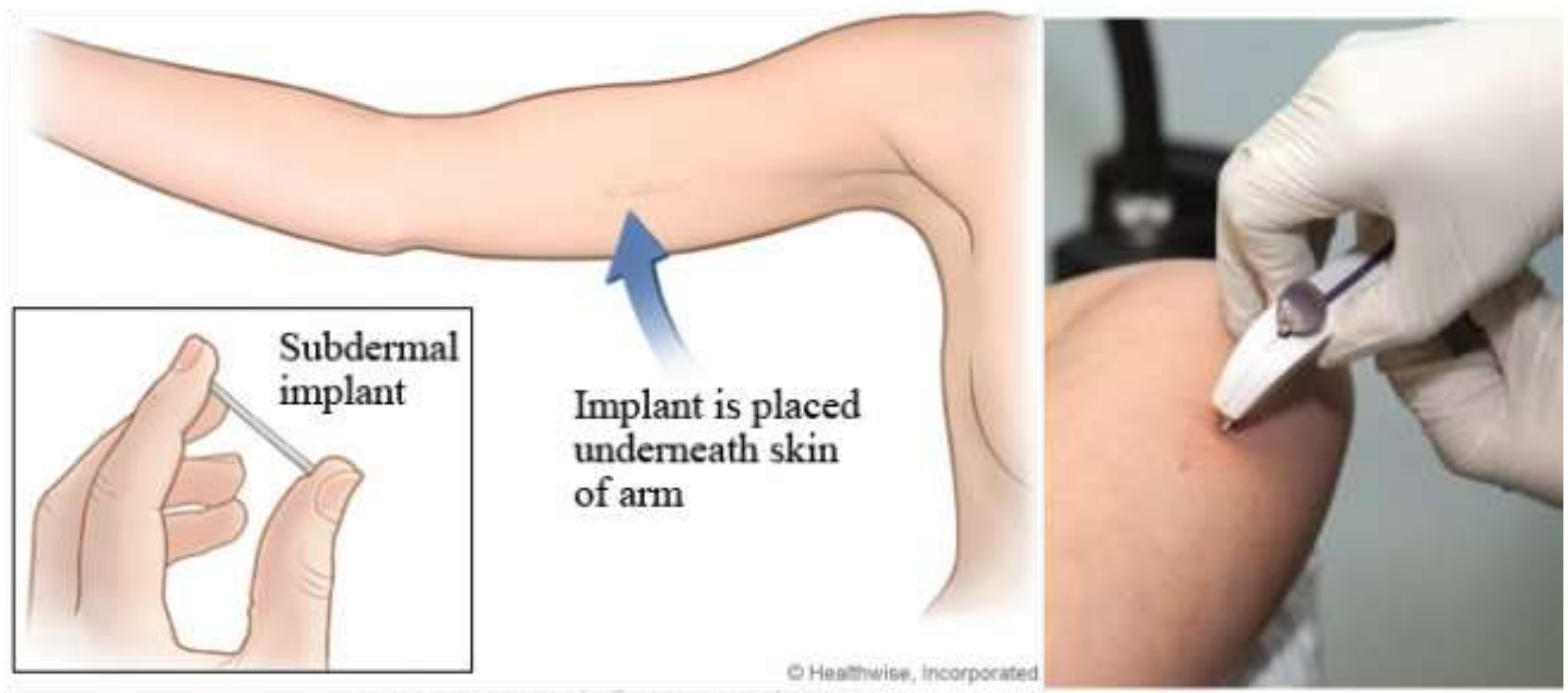

Infusion set
Continuous Glucose Monitor
Insulin pump
5.1

- Parenteral
  - Intravenous
  - Intramuscular
  - Subcutaneous
  - Intradermal
  - Transdermal

Subcutaneous administration.

2- Subcutaneous implantation

Implantation of pellets under the skin allows the release of the drug over a period of several weeks or months. (Steroid hormones).

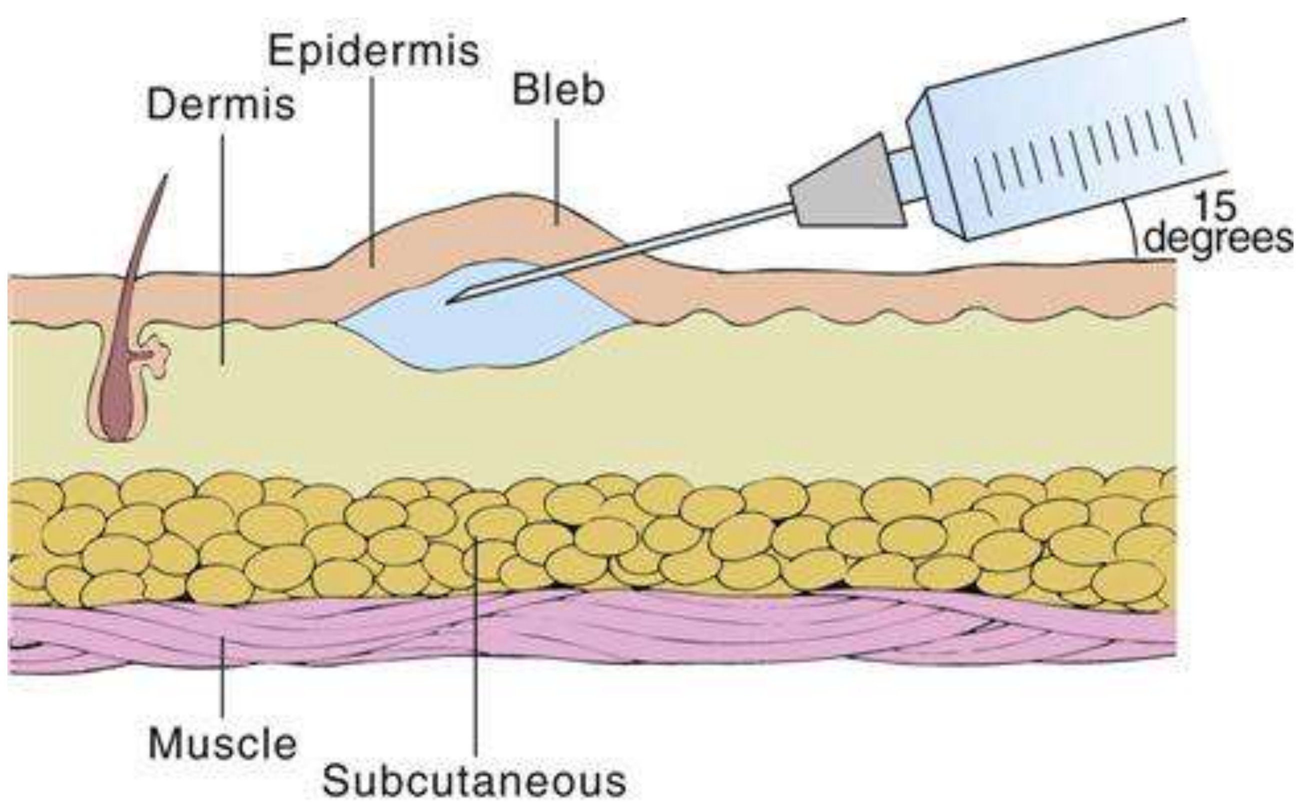


Routes of drug administration

- **Parenteral**
  - Intravenous
  - Intramuscular
  - Subcutaneous
  - Intradermal
  - Transdermal

Intradermal administration.

- The drug is injected into dermal tissue.
- This route is generally used for local effects (e.g., allergy testing such as Mantoux test for TB screening).
- Rarely used for systemic effects, due to poor distribution into the systemic circulation.
- The maximum injection volume is approximately **1 ml.**

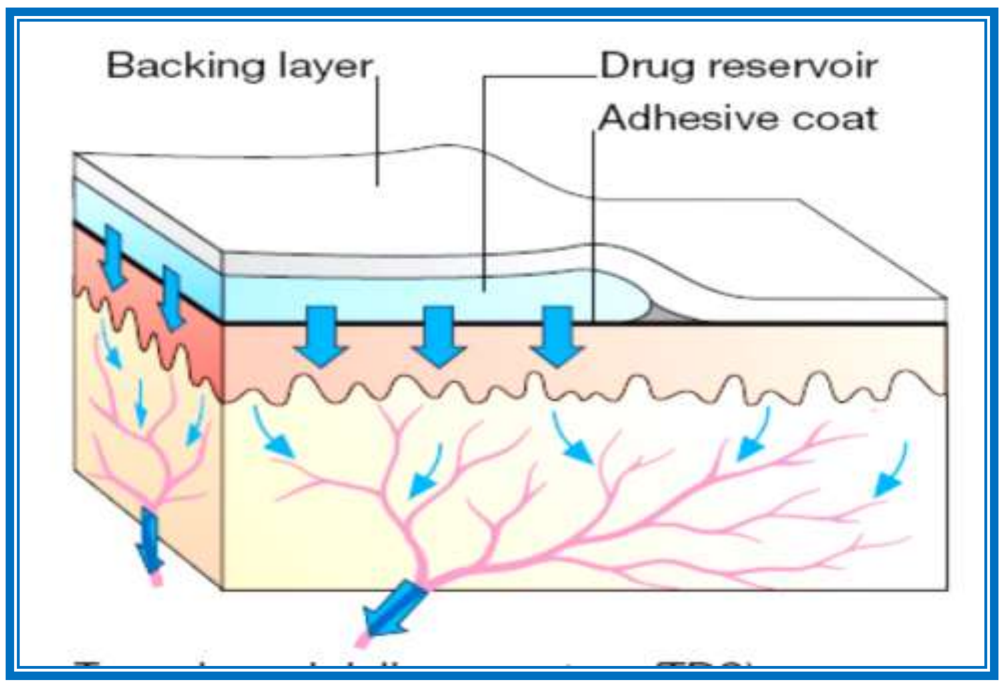


Routes of drug administration

- Parenteral
  - Intravenous
  - Intramuscular
  - Subcutaneous
  - Intradermal
  - Transdermal

Transdermal administration

This route of administration achieves systemic effects by application of drugs to the skin, usually via a transdermal patch .



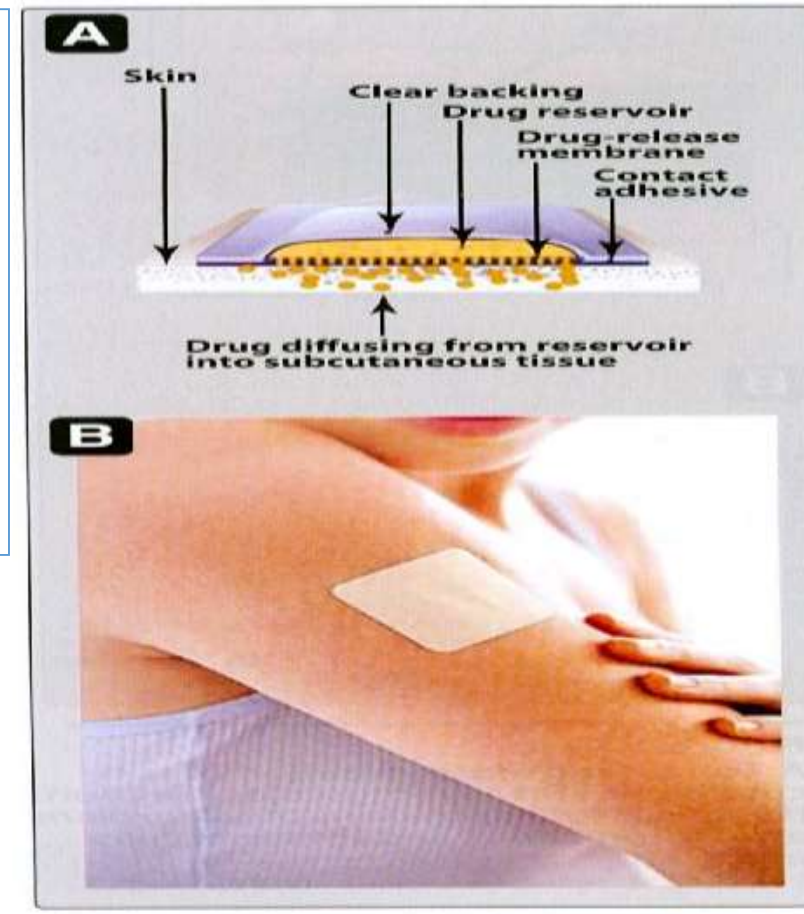


Figure 1.4
A. Schematic representation of a transdermal patch. B. Transdermal nicotine patch applied to arm.

•This route is most often used for the sustained delivery of drugs, such as the antianginal drug *nitroglycerin,* the antiemetic *scopolamine,* and nicotine transdermal patches, which are used to facilitate smoking cessation.

Routes of drug administration

- Enteral
  - Oral
  - Sublingual
  - Rectal

Oral administration.

▪The drug is swallowed (bioavailability <100%).

▪Some drugs must be administered in much larger doses orally than intravenously (e.g., propranolol requires an oral dose 20 – 40 times larger than its intravenous dose).

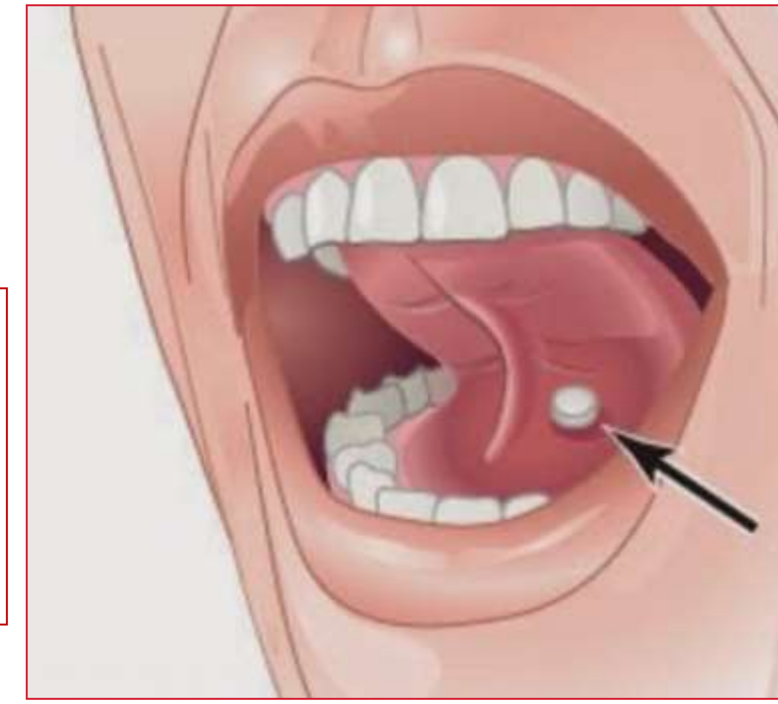
Advantages	Disadvantages
• Safe. • Convenient. • Economic.	• Delayed onset of action. • Not suitable for emergencies • Formation of complexes with food. • Destruction by digestive enzymes. • Inactivation by the liver (1st pass metabolism)

Routes of drug administration

- Enteral
  - Oral
  - Sublingual
  - Rectal

Sublingual administration.

The drug is administered by placement under the tongue (bioavailability is < 100% but often greater than with oral administration).



Advantages	Disadvantages
• **Rapid** absorption (due to the vascularity of sublingual tissues, and increased bioavailability, because sublingual circulation passes directly into the systemic circulation rather than into the portal circulation). • Absorbed directly to the systemic circulation. • Can be terminated by spitting out the tablet. • Avoid G.I.T enzymes and pH.	• not suitable for large volumes

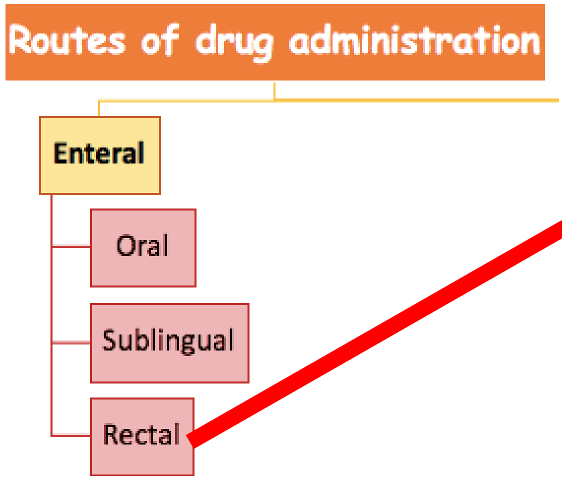
Routes of drug administration

- Enteral
  - Oral
  - Sublingual
  - Rectal

Buccal administration

-The drug is absorbed through the oral mucosa, after being placed between the cheek and the gums.

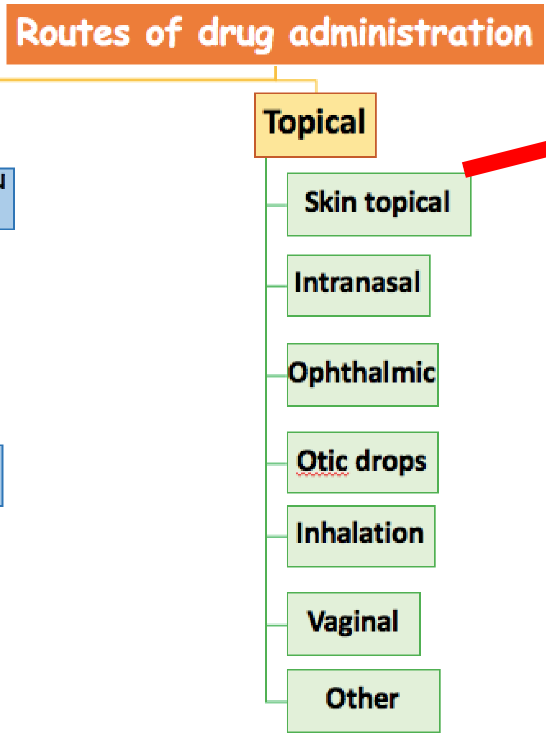
-Absorption is usually rapid, and bioavailability is similar to that of sublingual administration.



Rectal administration.

-The drug is administered rectally, usually in suppository form, or a liquid (bioavailability <100%).

Advantages	Disadvantages
• Most of the drug bypasses liver. • Avoids digestive enzymes. • Suitable for large volumes of fluids. • Can be used for administration when the patient is in vomiting or coma. • When cooperation is lacking (in children).	• Absorption is slow, erratic, and often incomplete. In addition, it may irritate the rectal mucosa.



Topical administration.

The drug is applied to the skin and absorbed into the systemic circulation after passing through the outermost epidermis.

Advantages

Disadvantages

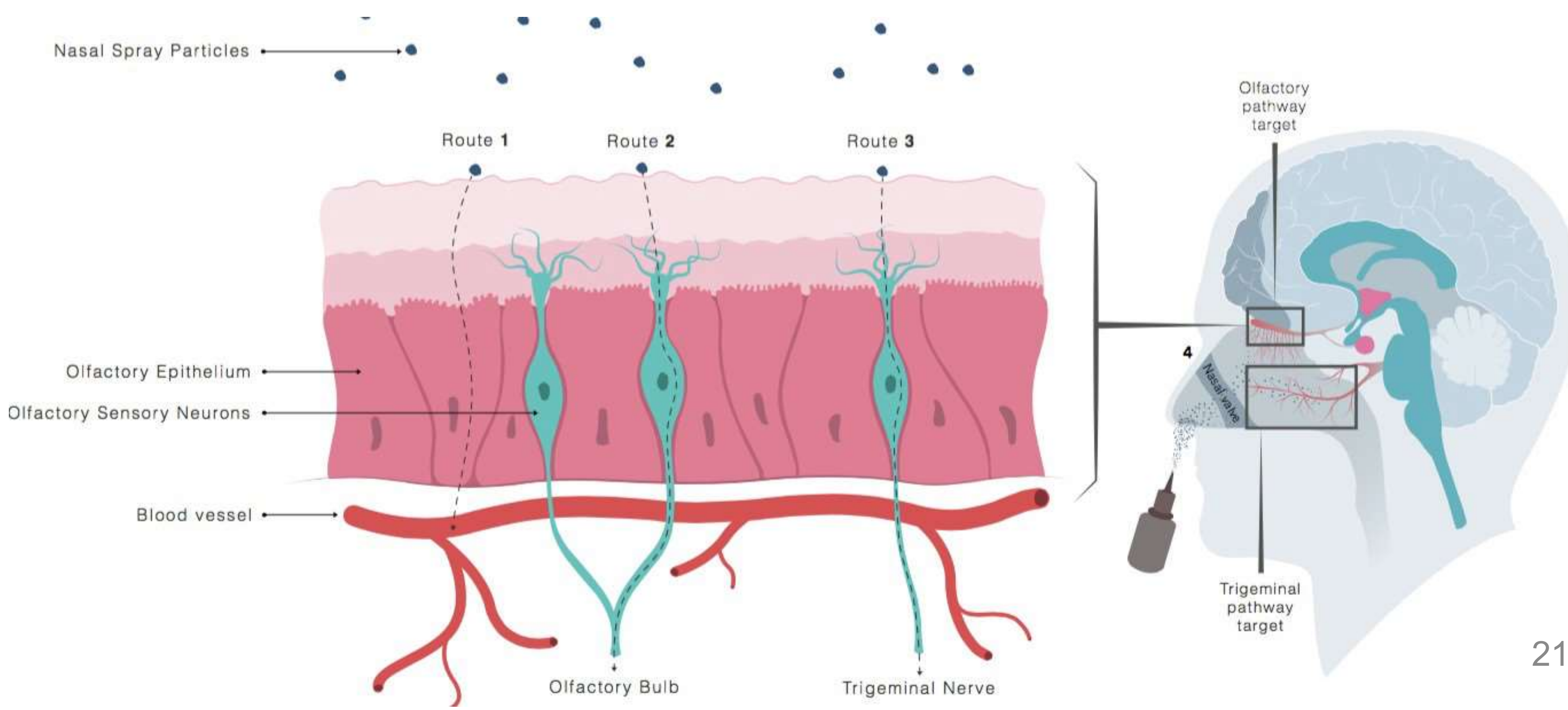
Advantages	Disadvantages
• long duration of action and the ability to remove the dosage form once it has been administered. • Provide high local concentration without systemic effect (low effect).	• Not suitable for irritant drugs.

Intranasal administration.

-The drug is inhaled nasally and then absorbed through the nasal mucosa.

-Onset is generally rapid, but bioavailability can be variable and dependent on the condition of nasal membranes.

-**Drugs may be administered for a local (e.g., phenylephrine) or systemic (e.g. desmopressin) effect**.



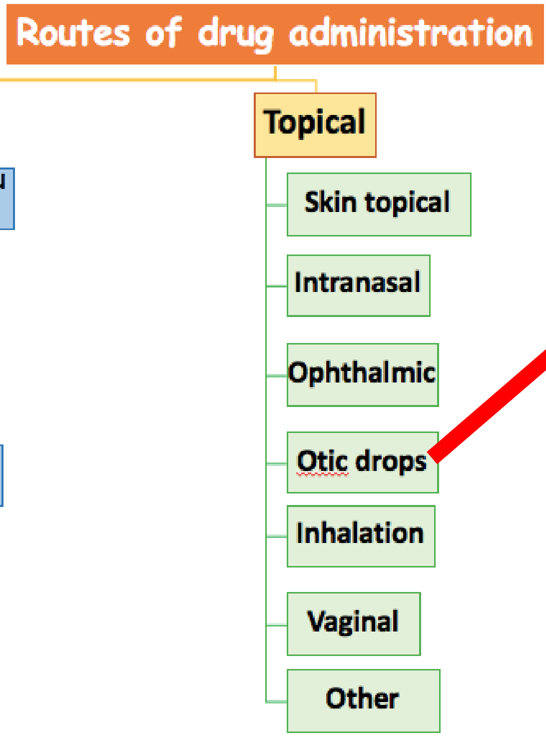
Routes of drug administration

- Topical
  - Skin topical
  - Intranasal
  - Ophthalmic
  - Otic drops
  - Inhalation
  - Vaginal
  - Other

Ophthalmic administration.

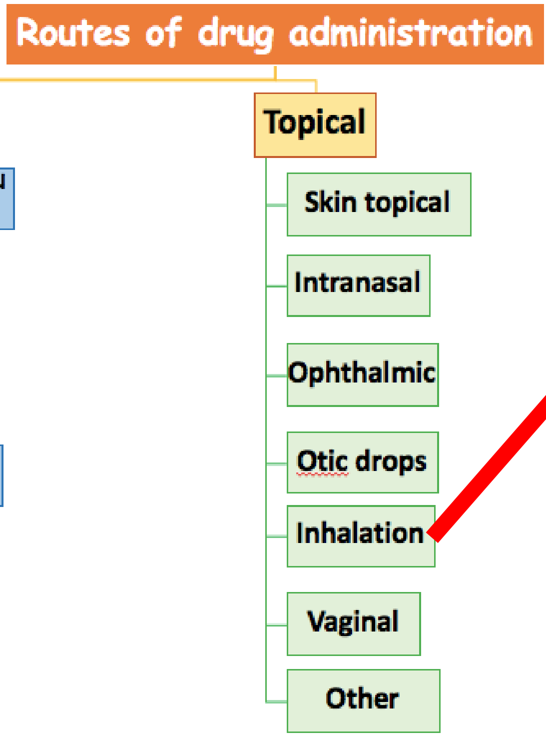
-The drug is administered onto the conjunctiva. The effect is usually local with systemic side effects occasionally occurring.

-Absorption into the central nervous system (CNS) is sometimes a problem because of the proximity of the eyes to the brain and the lipophilic nature of ophthalmically active medications.



Otic administration.

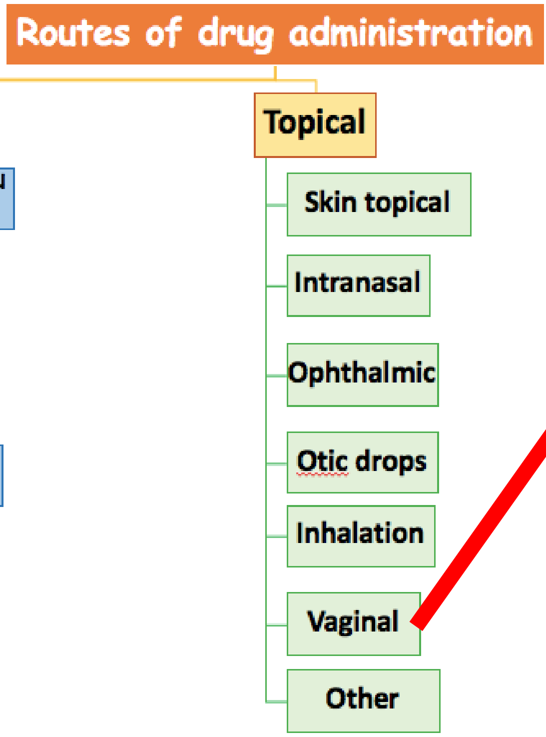
The drug is administered into the auditory canal, usually for local results, such as an anti-inflammatory or anti-infective effect.



Inhalation.

The drug is inhaled into the lungs, where the site of action may be local **(e.g., beclomethasone) or** **systemic** **(e.g., inhalational anesthetics).**

Advantages	Disadvantages
• Suitable for volatile compounds gases and aerosols. • Self administration is possible. • Can be used for local or systemic uses.	• Special apparatus is needed. • Not suitable for irritant drugs.

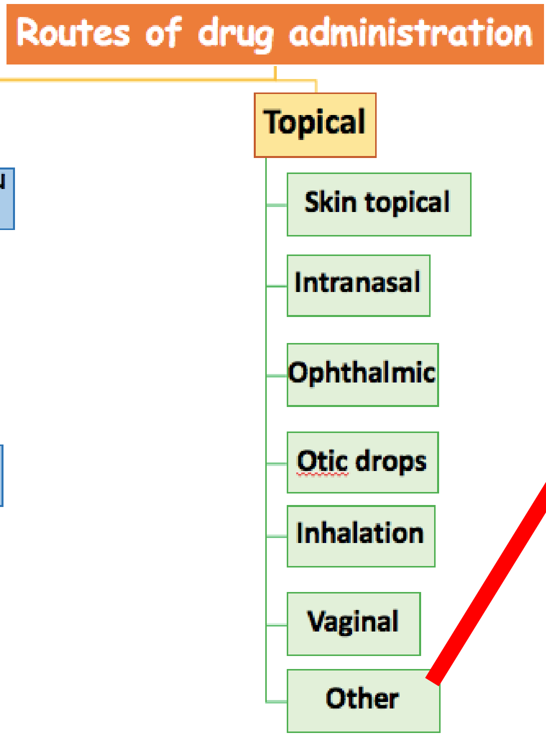


Vaginal administration.

-The drug is administered vaginally as a suppository, tablet, cream, or foam.

-Drug effects are usually local, but systemic absorption does occur through the vaginal mucosa.

- Systemic effects are usually side effects, rather than the desired therapeutic goal.



Other routes of drug administration

1- Local injection

- infiltration anesthesia

Injection of the drug around sensory nerve terminals.

- Block anesthesia

Injection of the drug around nerve trunk.

2- Intra-articular

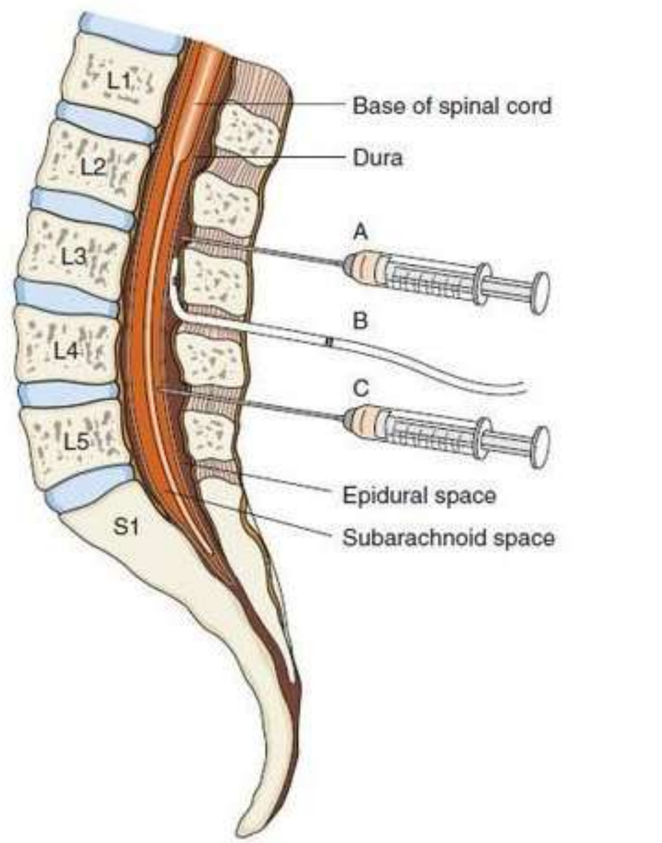
Intra-articular injection of corticosteroids in rheumatoid arthritis.

3- intrathecal.

Injection into subarachnoid space between L2 and L3.

4- Intracardiac (in emergencies)

5- Intra-arterial (for the preparation of arteriograms)



INJECTION OF SPINAL ANESTHESIA

(A) Epidural anesthesia, (B) epidural catheter, (C) spinal anesthesia

Thank you